Local Kafka Stack Quickstart: Confluent Platform, Kafka UI, and Apache Flink on Minikube

This tutorial provides a step-by-step guide to deploying the core components of the **Confluent Platform** with **Provectus' Kafka UI** and **Apache Flink** on a local Minikube Kubernetes cluster using **Confluent for Kubernetes (CFK)**, the **Provectus Kafka UI Helm chart**, and the **Flink Kubernetes Operator**.

By the end of this guide, you will have a fully functional **Confluent Platform** running in **KRaft mode (ZooKeeper-less Kafka)**, a browser-accessible **Kafka UI** for cluster visibility, and a **Flink session cluster** with its own UI, all running locally on Minikube. The guide also covers complete teardown procedures for each component.

The stack you will deploy includes:

- **Kafka (KRaft mode)** — brokers and controllers, no ZooKeeper
- **Schema Registry** — manages and enforces schemas for Kafka topics
- **Kafka Connect** — integrates Kafka with external systems
- **ksqlDB** — stream processing with SQL-like queries
- **REST Proxy** — provides a RESTful interface to Kafka
- **Control Center (C3)** — the Confluent monitoring UI
- **Provectus Kafka UI** — lightweight alternative UI for cluster inspection
- **Apache Flink** — session cluster via Flink Kubernetes Operator

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1.0 MacOS prerequisites setup

1.1 Prerequisites

Need a container driver running. Docker Desktop is a good choice:

```
brew install --cask docker
```

Then launch Docker Desktop and make sure it's running before proceeding.

Install `kubectl`:

```
brew install kubectl
```

1.2 Install Minikube

```
brew install minikube
```

If which minikube fails after install, relink it:

```
brew link --overwrite minikube
```

1.3 Verify Installation

```
minikube version
```

1.4 Start Minikube

CP is very memory and resource intensive, so we need to allocate more resources to the Minikube cluster. The default settings may not be sufficient for running CP components smoothly. We will allocate 6 CPU cores, 20GB of memory, and 50GB of disk space to ensure that the cluster can handle the workload effectively.

```
minikube start --driver=docker --cpus=6 --memory=20480 --disk-size=50g
```

- `--driver=docker`: Use Docker as the container runtime (instead of VirtualBox or Hyper-V).
- `--cpus=6`: Allocate 6 CPU cores to the Minikube cluster.
- `--memory=20480`: Allocate 20GB of memory to the Minikube cluster.
- `--disk-size=50g`: Allocate 50GB of disk space to the Minikube cluster.

Then verify that Minikube is running:

```
minikube status
kubectl get nodes
```

2.0 Startup the Confluent Operator (CFK)

2.1 Create the Confluent namespace and set it as the default context for `kubectl`

```
kubectl create namespace confluent
kubectl config set-context --current --namespace=confluent
```

2.2 Add the Confluent Helm repo and install the Confluent Operator

```
helm repo add confluentinc https://packages.confluent.io/helm
helm repo update
helm upgrade --install confluent-operator confluentinc/confluent-for-
kubernetes --namespace confluent
```

2.3 Verify the Operator is running

```
kubectl get pods -n confluent
```

If all goes well you should see the following output:

```
NAME                                  READY   STATUS    RESTARTS   AGE
confluent-operator-66b887b979-frj42   1/1     Running   0          15s
```

The CFK operater is now running and ready to manage Confluent Platform resources in the cluster.

3.0 Deploy KRaft broker and controller with CFK

3.1 Set the TUTORIAL_HOME environment variable

```
export
TUTORIAL_HOME="https://raw.githubusercontent.com/confluentinc/confluent-kubernetes-examples/master/quickstart-deploy/kraft-quickstart"
```

The `TUTORIAL_HOME` environment variable is being set to a raw GitHub URL that points to the directory containing the Kubernetes manifest for deploying Confluent Platform in KRaft mode. This allows us to reference the manifest file directly from that URL in the next step when we apply it with `kubectl`.

Sets an environment variable pointing to a raw GitHub URL for Confluent's official Kubernetes examples repo — specifically the KRaft quickstart directory. KRaft is Kafka's ZooKeeper-free mode (Kafka Raft metadata).

> **To browse the manifest file directly, you can visit:** https://github.com/confluentinc/confluent-kubernetes-examples/tree/master/quickstart-deploy/kraft-quickstart

3.2 Apply the Confluent Platform manifest

```
kubectl apply -f $TUTORIAL_HOME/confluent-platform-c3++.yaml
```

> The `raw.githubusercontent.com` URL works when you append a specific filename to it, allowing you to directly access the raw content of that file. When you use `kubectl apply -f` with a URL, `kubectl` can fetch the YAML manifest directly from that URL and apply it to your Kubernetes cluster without needing to download the file manually first. This is a convenient way to deploy Kubernetes resources directly from a GitHub repository.

Applies a Kubernetes manifest file directly from that GitHub URL, `kubectl` can fetch and apply remote YAML files without downloading them first. The c3++ in the filename likely refers to Control Center (C3) with additional/enhanced configuration (the ++ suggesting an extended or "plus" variant).

> This is spinning up the Confluent Platform on Kubernetes using the Confluent Operator (CFK - Confluent for Kubernetes) in KRaft mode. It would typically deploy:
>
> - Kraft Controllers
> - Kafka Brokers
> - Schema Registry
> - Kafka Connect

> - ksqlDB
> - REST Proxy
> - Control Center (C3) — the Confluent monitoring UI

3.3 Verify the platform is running

Watch the pods in the confluent namespace until all components are up and running (this will take a few minutes):

```
kubectl get pods -n confluent -w
```

You're looking for all pods to be in the "Running" state.

Final status checker for Kubernetes Operators:

```
kubectl get pods -n confluent
```

You should see all pods at `1/1` or `3/3 Running`. Then open the Control Center UI to verify the platform is healthy.

Open Control Center:

```
kubectl port-forward -n confluent controlcenter-0 9021:9021
```

3.4 Access Control Center

Go to the Control Center UI dashboard:

```
http://localhost:9021/
```

4.0 Teardown the CP Core Components

To clean up the resources created by this tutorial, you can delete the Confluent Platform deployment and the Confluent Operator:

```
kubectl delete -f $TUTORIAL_HOME/confluent-platform-c3++.yaml
helm uninstall confluent-operator -n confluent
kubectl delete namespace confluent
```

5.0 Deploy Provectus' Kafka UI

5.1 Add the Helm repo

```
helm repo add kafka-ui https://provectus.github.io/kafka-ui-charts
helm repo update
```

5.2 Install Kafka UI

```
helm upgrade --install kafka-ui kafka-ui/kafka-ui \
  --namespace confluent \
  --set yamlApplicationConfig.kafka.clusters[0].name="confluent" \
  --set
yamlApplicationConfig.kafka.clusters[0].bootstrapServers="kafka:9092" \
  --set
yamlApplicationConfig.kafka.clusters[0].schemaRegistry="http://schemaregis
try:8081" \
  --set
yamlApplicationConfig.kafka.clusters[0].kafkaConnect[0].name="connect" \
  --set
yamlApplicationConfig.kafka.clusters[0].kafkaConnect[0].address="http://co
nnect:8083" \
  --set yamlApplicationConfig.auth.type="DISABLED" \
  --set yamlApplicationConfig.management.health.ldap.enabled="false"
```

The service names (`kafka`, `schemaregistry`, `connect`) are the internal Kubernetes DNS names assigned by CFK, they only resolve from within the cluster, which is why Kafka UI runs inside the same namespace.

5.3 Verify the installation

```
kubectl get pods -n confluent | grep kafka-ui
```

Wait until the pod shows `Running` before port-forwarding.

5.4 Open the UI

```
kubectl port-forward -n confluent svc/kafka-ui 8080:80
```

Then open `http://localhost:8080` in your browser. Note the asymmetry: the service listens on port `80` internally, forwarded to `8080` locally.

6.0 Teardown Kafka UI

To remove Kafka UI from the cluster, run:

```
helm uninstall kafka-ui -n confluent
```

7.0 Deploy Apache Flink on Kubernetes with the Flink Operator

Set these shell variables once at the top of your terminal session to avoid hardcoding versions in the commands below:

```
export CERT_MANAGER_VER="v1.18.2"
export FLINK_OPERATOR_VER="1.130.0"
```

7.1 Install `cert-manager` manifest

The Confluent Flink Kubernetes Operator uses cert-manager webhooks for TLS certificate injection. cert-manager must be fully ready **before** the operator is installed:

```
kubectl apply -f https://github.com/jetstack/cert-manager/releases/download/${CERT_MANAGER_VER}/cert-manager.yaml
```

> This creates the `cert-manager` namespace and installs three deployments: `cert-manager`, `cert-manager-cainjector`, and `cert-manager-webhook`.

7.2 Wait for cert-manager to be ready

Then wait for all three components to be ready before proceeding, the Flink operator webhook will fail silently if `cert-manager` isn't fully up:

```
# Core controller
kubectl wait --for=condition=ready pod -l app=cert-manager -n cert-manager --timeout=120s

# CA injector (patches webhook CA bundles)
kubectl wait --for=condition=ready pod -l app=cainjector -n cert-manager --timeout=120s

# Webhook (validates Certificate / Issuer CRs)
kubectl wait --for=condition=ready pod -l app=webhook    -n cert-manager --timeout=120s
```

7.3 Verify cert-manager is running

```
kubectl get pods -n cert-manager
```

7.4 Install the Confluent Flink Kubernetes Operator

The Flink Operator is available on the Confluent Helm repo, so we can install it with `helm`:

7.4.1 Add the Confluent Helm repo

```
helm repo add confluentinc https://packages.confluent.io/helm 2>/dev/null
|| true
helm repo update
```

Confirm the chart is available:

```
helm search repo confluentinc/flink-kubernetes-operator --versions | head
-5
```

7.4.2 Install the operator

```
helm upgrade --install cp-flink-kubernetes-operator \
  confluentinc/flink-kubernetes-operator \
  --version "~${FLINK_OPERATOR_VER}" \
  --namespace confluent \
  --set watchNamespaces="{confluent}" \
  --set webhook.create=false
```

Flag	Value	Reason
--version	"~${FLINK_OPERATOR_VER}"	Tilde constraint allows patch-level upgrades only
watchNamespaces	"{confluent}"	Scopes operator to the confluent namespace; prevents cluster-wide RBAC grants
webhook.create	false	Disables the operator's own admission webhook (cert-manager is handling TLS for CMF; the operator webhook is not required for session-cluster mode)

7.4.3 Wait for the operator pod

```
kubectl wait --for=condition=ready pod -l app.kubernetes.io/name=flink-
kubernetes-operator -n confluent --timeout=180s
```

7.4.4 Verify

```
# Pod status
kubectl get pods -n confluent | grep -E "flink-kubernetes-operator"

# Helm release
helm list -n confluent | grep cp-flink-kubernetes-operator

# CRDs installed by the operator
kubectl get crds | grep flink.apache.org
```

8.0 Deploy Confluent Manager for Apache Flink (CMF) on Kubernetes

Set these shell variables once at the top of your terminal session to avoid hardcoding versions in the commands below:

```
export CMF_VER="2.1.0"
export CMF_ENV_NAME="dev-local"
export CMF_LOCAL_PORT="18080"   # ephemeral port used for port-forward
during API calls
export CMF_OPEN_PORT="8080"     # port used when browsing the REST API
interactively
```

8.1 Add (or update) the Confluent Helm repo

```
helm repo add confluentinc https://packages.confluent.io/helm 2>/dev/null
|| true
helm repo update
```

Confirm the chart is available:

```
helm search repo confluentinc/confluent-manager-for-apache-flink --
versions | head -5
```

8.2 Install the CMF chart

```
helm upgrade --install cmf confluentinc/confluent-manager-for-apache-flink
--version "~${CMF_VER}" --namespace confluent --set
cmf.sql.production=false
```

Flag	Value	Reason
--version	"~${CMF_VER}"	Tilde allows patch-level upgrades; pins the minor version

Flag	Value	Reason
cmf.sql.production	false	Enables non-production SQL mode required for local / dev deployments; Flink SQL statements will not enforce production-grade resource quotas

8.3 Wait for the CMF pod

```
kubectl wait --for=condition=ready pod -l
app.kubernetes.io/name=confluent-manager-for-apache-flink -n confluent --
timeout=180s
```

8.4 Verify

```
# Pod status
kubectl get pods -n confluent -l app.kubernetes.io/name=confluent-manager-
for-apache-flink

# Helm release
helm list -n confluent | grep cmf

# CMF service
kubectl get svc cmf-service -n confluent
```

8.1 Remove Flink cluster

```
kubectl delete flinkdeployment flink-basic -n confluent --ignore-not-
found=true
```

8.2 Remove Flink Operator

```
helm uninstall flink-kubernetes-operator -n confluent
```

8.3 Remove cert-manager

```
kubectl delete -f https://github.com/jetstack/cert-
manager/releases/download/${CERT_MANAGER_VER}/cert-manager.yaml --ignore-
not-found=true
```

9.0 Glossary

Term	Description
CFK	Confluent for Kubernetes, a set of Kubernetes Operators for deploying and managing Confluent Platform on Kubernetes.
CRD	Custom Resource Definition, a way to extend Kubernetes capabilities by defining custom resources.
Operator	A method of packaging, deploying, and managing a Kubernetes application.
Controller	A control loop that watches the state of your cluster and makes or requests changes as needed.
Reconciliation	The process of ensuring that the current state of the cluster matches the desired state defined by a resource.
Finalizer	A mechanism to perform cleanup before a resource is deleted.
Webhook	A way to extend Kubernetes API by intercepting API requests and modifying them or validating them.
CR	Custom Resource, an instance of a CRD that represents a specific configuration or state in the cluster.

10.0 Kubernetes Nautical Theme

The nautical theme runs throughout the Kubernetes ecosystem:

Term	Description
Kubernetes	Helmsman/pilot
Helm	The ship's wheel
Charts	Nautical maps (Helm packages are called "charts")
Harbor	A container registry (like a port where ships dock)
Fleet	A multi-cluster management tool